Visual-RSI: Can Agents Turn Visual Evidence into Better Training Data?

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Abstract

Can an agent use a visual model's failures to design more effective training data? Visual-RSI tests this question on charts, 2D/3D geometry, flow diagrams, and web interfaces. The agent identifies these failures and builds new training data. We then retrain the model from scratch to measure the result. The main experiment evaluates seven inference models across four research harnesses, yielding 28 configurations. Each run makes five attempts from the same starting model; we keep the research history but retrain from the initial weights every time. A four-condition ablation compares four diagnostic-feedback settings: scores only, text errors, error images, and verified evidence regions. Separate studies test research history, visual correspondence, reasoning effort, and individual operations. One case illustrates a complete sequence. We implement verifiable chart construction and show positive, null, and adverse outcomes. ***All model results are simulated rather than empirical findings.*** *Local checks validate data construction only. The protocol reuses one evaluation set. Real validation requires independent execution.*

1. Introduction

Visual benchmarks typically ask whether a model can answer a question about a chart, reason about a diagram, or identify a screen element [7, 17, 21]. We ask a different question: can an agent study another model's visual failures and construct training data that improves that model? We do not evaluate the Researcher on its own visual answer. Instead, we measure a separate target vision-language model (VLM) after controlled training.

This framing makes data design itself an object of study. A wrong chart answer could come from an arithmetic error, a misread axis, or a broken legend-to-series link. Each error calls for a different fix. For example, changing a bar's value requires a corresponding update to the answer, whereas swapping colors should leave the answer unchanged. The research problem connects each observed failure to a hypothesis, a data intervention, and evidence from a newly trained model.

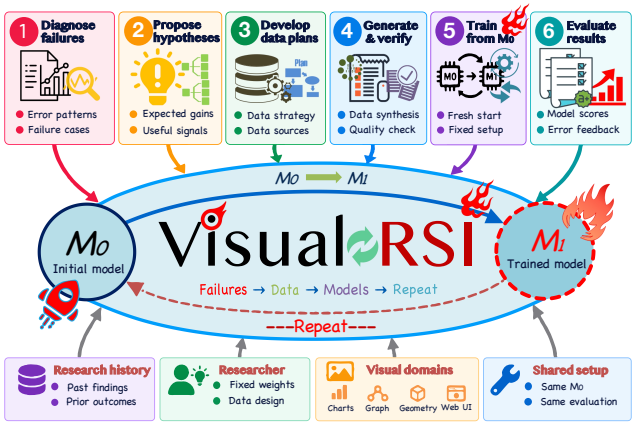


Figure 1. Visual data research: failure evidence guides training data for a separate vision-language model.

RSIBench-Data studies data research under a shared post-training stack [23], while DataEnvGym provides a related teacher–student environment for visual tasks [18]. We extend a common-initialization loop with verifiable visual interventions, nested diagnostic feedback, and independently configurable models and harnesses.

A model supplies inference, whereas a harness organizes context, tools, and research state. Evaluating four harnesses with seven models lets us test matched configurations beyond vendor-native pairs. Separately, we vary diagnostic feedback—scores only, text errors, error images, or evidence regions—to measure the value of each type of information. Figure 2 introduces the task, with Figure 3 presenting the execution protocol. Our contributions are threefold:

- **Verifiable visual interventions:** We modify charts and diagrams while automatically updating their answers and evidence regions. A chart prototype implements local verification rules.
- **A common-initialization research environment:** We evaluate seven models with four harnesses, yielding 28

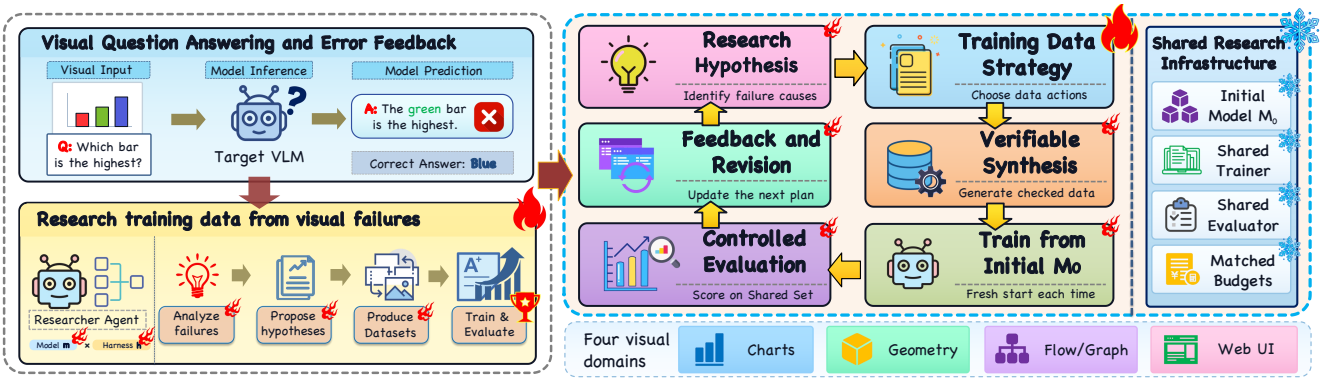


Figure 2. From visual task solving to data research. Left: Answering visual questions versus designing training data for a separate target VLM. Right: Diagnose errors, construct verifiable data, train candidates from the same M_0, and evaluate them.

configurations. Every configuration starts from the same initial model.

- **Controlled comparisons:** We examine which information helps by varying diagnostic feedback, research history, verified evidence, and data operations.

The model study uses synthetic data to illustrate this design. Its results therefore do not support empirical claims about agent rankings or improvement.

Table 1 and Figure 4 report final target performance and five-attempt progress across configurations; Table 2 focuses on diagnostic feedback; in parallel, the controlled-studies section covers research history, visual correspondence, reasoning effort, data operations, and agent-free baselines; Figure 5 illustrates one research sequence.

2. Related Work

Visual benchmarks measure a model's ability to interpret charts, diagrams and interfaces [7, 17, 21]. We ask a different question: can an agent see where a model fails and build better training data?

Adaptive data research. RSIBench-Data studies agent-driven training-data research, while DataEnvGym formulates teaching as sequential data-generation decisions and includes visual tasks [18, 23]. DataComp-LM, CurationBench and broader post-training studies examine other parts of the learning pipeline [5, 15, 19, 26]. We study visual data interventions with configurable agents under shared training and evaluation services.

Verifiable visual synthesis. Socratic-Geo and VISA use solver or verifier feedback to guide visual data generation [12, 31]. Code-based synthesis and structured visual datasets provide checkable supervision [13, 14, 24, 29], while controlled edits probe visual dependence [2, 3, 9]. We apply these construction principles across experiments and test whether a Researcher can select useful interventions.

Agent systems and self-improvement. Self-Instruct and STaR improve models through generated training data; Self-Refine instead revises outputs at inference time [20, 28, 30]. On-policy Data Evolution uses a visual-native harness to refine training data through model rollouts [11]. Our Researcher's weights stay fixed; only its history and data strategy evolve.

3. Visual-RSI Framework

We define the visual data research task, introduce verifiable interventions and modular agent components, then describe the five-attempt research protocol.

3.1. Tasks and Verifiable Data Interventions

Four domains cover charts, 2D/3D geometry, flow/graph reasoning and web interfaces. Training materials include structured tables and plot specifications, constrained constructions or scenes, attributed graphs, and local component trees. Each domain has predefined task and answer rules. The 2D and 3D geometry tracks share one training job. We measure the target VLM's performance after training it on the generated data. For web-interface tasks, we evaluate predictions on static screenshots rather than completion of live browsing tasks.

The construction engine takes a source state, rendering configuration, and question program, then produces an image and an answer, with evidence regions included only when verified. Permitted operations include question construction, value/layout edits, binding permutations and checked negatives (catalogue in Appendix F).

Each operation must preserve or recompute the answer and its supporting evidence. For example, when a question asks for a bar's value, changing that value from 6 to 9 requires updating the answer to 9. By contrast, a type-safe

Table 1. Full-feedback, high-effort results (three-repeat means). Scores: %; gains: pp. ΔM_0: five-track mean gain; Transfer and General: four-domain means (Appendix M); Reach: trajectories reaching $+5$ pp (out of 12); Time: active h/trajectory.

Harness	Model	Chart	Geo^{2D}	Geo^{3D}	Flow	Web	ΔM_0	Reach	Transfer	General	Time
Target M_0		52.4	43.9	43.9	48.8	41.2	—	—	—	—	—
Codex	Opus 5.5	64.6	50.9	52.2	60.4	50.4	+9.7	11/12	+7.2	-0.9	8.7
	GPT-6.1 Sol	65.0	47.1	45.8	53.5	46.7	+5.6	9/12	+4.3	-0.5	8.0
	DS V4.1 Flash	62.6	54.9	53.8	56.4	45.1	+8.5	11/12	+5.6	-1.0	7.4
	Kimi K3	65.8	49.8	50.1	**63.4**	45.2	+8.8	9/12	+6.6	-0.6	7.9
	GLM-5.3 Flash	60.2	50.2	46.8	56.9	47.7	+6.3	10/12	+4.5	**-0.2**	8.5
	Fable 5.1	63.1	50.3	50.3	61.8	50.1	+9.1	**12/12**	+6.9	**-0.4**	8.0
	Sonnet 5.5	65.0	51.8	50.8	58.9	48.7	+9.0	**12/12**	+6.5	-1.0	8.0
Claude Code	Opus 5.5	63.3	51.7	52.4	63.2	49.1	+9.9	12/12	+6.9	-0.4	8.6
	GPT-6.1 Sol	59.2	53.3	52.6	57.9	45.6	+7.7	10/12	+5.2	-0.5	7.8
	DS V4.1 Flash	62.0	54.4	51.6	56.9	**50.7**	+9.1	11/12	+6.1	-0.6	7.3
	Kimi K3	62.3	56.4	**60.8**	60.9	47.3	**+11.5**	11/12	+6.7	-0.6	7.4
	GLM-5.3 Flash	**66.2**	48.9	51.7	60.3	46.8	+8.7	10/12	+6.1	-0.7	8.6
	Fable 5.1	61.1	51.8	53.3	**64.3**	48.3	+9.7	11/12	+7.1	-0.8	7.9
	Sonnet 5.5	62.7	49.6	52.9	58.5	47.6	+8.2	**12/12**	+6.0	-0.8	8.0
Qwen Code	Opus 5.5	62.5	**57.3**	52.9	59.1	48.4	+10.0	11/12	+6.5	-0.7	**9.0**
	GPT-6.1 Sol	63.5	52.3	52.4	61.3	**50.6**	+10.0	**12/12**	**+7.4**	-0.4	7.9
	DS V4.1 Flash	**66.3**	51.8	50.2	58.9	46.3	+8.7	11/12	+6.1	-0.6	7.4
	Kimi K3	65.4	**58.2**	**56.9**	61.4	49.9	**+12.3**	**12/12**	**+8.0**	-0.7	7.5
	GLM-5.3 Flash	59.0	48.2	50.6	56.3	48.4	+6.5	11/12	+4.3	-0.7	8.5
	Fable 5.1	65.1	55.0	55.3	59.3	49.8	+10.9	**12/12**	+7.2	-0.7	8.2
	Sonnet 5.5	61.9	51.8	51.8	62.9	48.9	+9.4	10/12	+6.9	-0.9	8.0
OpenCode	Opus 5.5	63.9	50.7	51.6	59.1	50.1	+9.0	**12/12**	+6.6	-0.6	**9.1**
	GPT-6.1 Sol	62.1	55.2	54.1	57.6	43.9	+8.5	10/12	+5.6	-0.8	8.0
	DS V4.1 Flash	63.6	50.0	46.4	58.0	43.7	+6.3	9/12	+4.7	-0.5	7.4
	Kimi K3	63.1	54.0	51.1	56.6	46.2	+8.1	9/12	+5.1	-0.5	7.7
	GLM-5.3 Flash	63.0	51.6	51.6	57.6	48.8	+8.5	**12/12**	+6.1	-0.8	8.7
	Fable 5.1	64.3	51.2	49.0	58.9	45.6	+7.8	10/12	+5.7	-0.6	8.0
	Sonnet 5.5	64.2	49.3	51.4	59.0	46.9	+8.1	11/12	+5.7	-0.6	7.8

color permutation can leave the answer unchanged. Each contrast pair retains the original and modified source states and records which factor changed, allowing the supervision to be audited.

Training examples must have valid schemas, executable answer programs, legible renderings, and consistent edits, and must pass checks for duplication and evaluation contamination. Training examples must be traceable to permitted source snapshots and must not be derived from evaluation images or paraphrased evaluation answers.

Our local chart prototype implements these checks for lookup and difference questions. Fourteen authored fixtures pass 171 checks; 24 matched operation-on/off pairs pass 536 checks. These checks verify answer recomputation and evidence-region consistency after chart edits. They do not measure the Researcher's diagnostic ability or the target model's learning. Verification details are provided in Appendix G.

3.2. Modular Researchers and Shared Services

A Researcher $\mathcal{R}_{m,h}$ combines inference model m with harness h, which manages context, tools and research history. We evaluate different combinations of inference models and harnesses across trajectories, keeping each combination fixed within its trajectory. During research, we keep the inference model's weights fixed and train only the target VLM on the generated data. For each task domain d, shared services provide seed materials, an online discovery gateway, a frozen evaluation collection E_d, the initial target model M_0, a training recipe c, and a resource budget.

All harnesses access the same construction, verification, training and evaluation capabilities under a shared task contract described in Appendix E. We document each harness's policies for managing context and executing tasks to make differences between systems explicit. Provider adapters may translate message and tool formats, but must not silently substitute models, introduce additional reasoning agents, or expose restricted evaluation data. We keep the shared services fixed across configurations to compare how different Researchers design training data.

For each accepted source snapshot, the discovery gateway records its URL, timestamp, content hash, and license, and tracks which training examples use it. The Researcher may search for training materials based on general skill gaps suggested by observed failures, but must exclude evaluation questions, answers, images, and identifiers from search queries and training supervision. We treat retrieved content as source material and do not allow it to override the tools

or resource budgets. As research progresses, online discovery allows the Researcher to retrieve new materials and add accepted snapshots to the source collection. Appendix G describes source safeguards and verification procedures.

3.3. Five Attempts from a Common Initial Model

Initial feedback F_0^a comes from evaluating M_0 on E_d and includes only the information allowed under feedback condition a. At attempt t, the Researcher uses source snapshots $S_{d,t}$ and permitted research history H_{t-1}^a to propose a data plan π_t and construct a complete dataset D_t. Candidate training always restarts from the common initialization:

$$\begin{aligned}(\pi_t, D_t) &= \mathcal{R}_{m,h}(S_{d,t}, F_0^a, H_{t-1}^a),\\ M_t &= \mathrm{Train}(M_0, D_t; c).\end{aligned} \tag{1}$$

The Researcher can add newly accepted online snapshots to the source collection $S_{d,t}$ as research progresses. Evaluation produces the score $s_t = \mathrm{Eval}(M_t, E_d)$ and feedback allowed under condition a. The harness updates the research history with each attempt's data plan, dataset, feedback, and score.

When a candidate performs worse than the current best, we still update the research history with that candidate's own feedback. Each trajectory has five attempts. A failed data proposal or training job still consumes an attempt and its associated resources. A documented recovery from an infrastructure interruption continues the same attempt.

We select the valid candidate with the highest unrounded domain score, choosing the earliest attempt in a tie and reporting a missing result if the valid set $\mathcal{V}$ is empty.

$$t^\star = \min \arg\max_{t \in \mathcal{V}} s_t. \tag{2}$$

If the selected candidate scores below the initial model M_0, we report a negative gain. Regardless of the gain, we freeze the selected checkpoint and evaluate it once more on the same evaluation collection E_d. We record the resulting score without updating the Researcher's history or reopening candidate selection.

The final score reflects the performance of the selected model on the tasks reused during research. To assess generalization beyond these tasks, we also evaluate the model on a frozen transfer collection and a general visual evaluation suite, as reported in Table 1.

3.4. Diagnostic Information for Data Research

The main study uses the **evidence regions** condition, which provides the most detailed feedback. To examine how different levels of diagnostic information affect data research, we compare four conditions, each fixed from the initial diagnosis through all five attempts.

In the **scores only** condition, the Researcher receives evaluation scores. The **textual errors** condition additionally provides the questions the target model answered incorrectly, its predictions, and the reference answers. The **error images** condition retains this information and includes the original images for those questions. The **evidence regions** condition retains all of the above and adds verified regions indicating which parts of the images are relevant to the questions.

We use the same sampling rules, limits on feedback size, and data-construction tools in all four conditions. The error cases may still change across attempts because candidate models can fail on different evaluation items. If a sampled case has no available evidence regions, we keep it in the feedback without region annotations.

Reference answers may already contain coordinates, so this comparison tests whether adding evidence regions further improves the performance of the target model. The Researcher may view training-source images in every condition, but may not use logs or tools to access evaluation images that the condition does not provide.

4. Experiments

We present the experimental setup, main results, ablations of diagnostic feedback and research processes, and a case study.

4.1. Experimental Setup

We pair each of the four research harnesses Codex, Claude Code, Qwen Code, and OpenCode with seven inference models: Opus, GPT, DeepSeek, Kimi, GLM, Fable, and Sonnet. Each of the 28 configurations designs training data for the same target model, Qwen3-VL-4B-Instruct. Across all configurations, we train this model with language-side LoRA [1, 10] and use the same tools for data generation, training, and evaluation. Table 1 lists the model versions. For the main study, we use the evidence regions condition and set reasoning effort to high.

We use ChartQA, ChartQAPro, and PlotQA for chart tasks; GeomVerse, MathVerse, and BabyVision for geometry; FlowVQA, FlowLearn, and AI2D for flow and graph tasks; and ScreenSpot, WebSRC, and Mind2Web for web interfaces [4, 6–8, 16, 17, 21, 22, 24, 25, 27, 32]. Each domain has 600 evaluation items, and we evaluate the same trained model on the eligible 2D and 3D subsets for geometry. We give equal weight to all five evaluation tracks when computing the mean. Before research begins, we fix how each dataset is converted into evaluation tasks, together with the reference answers and available evidence regions.

In each attempt, we train a candidate model on 1,000 verified records for one epoch, using LoRA rank 8, a learning rate of 10^{-4}, and an effective batch size of 16. After evaluation, the feedback includes at most 20 sampled error cases. For each model–harness configuration and domain, we use three trajectory repeats per condition, with seeds 17, 29, and 43. We hold the model, harness, domain, and repeat

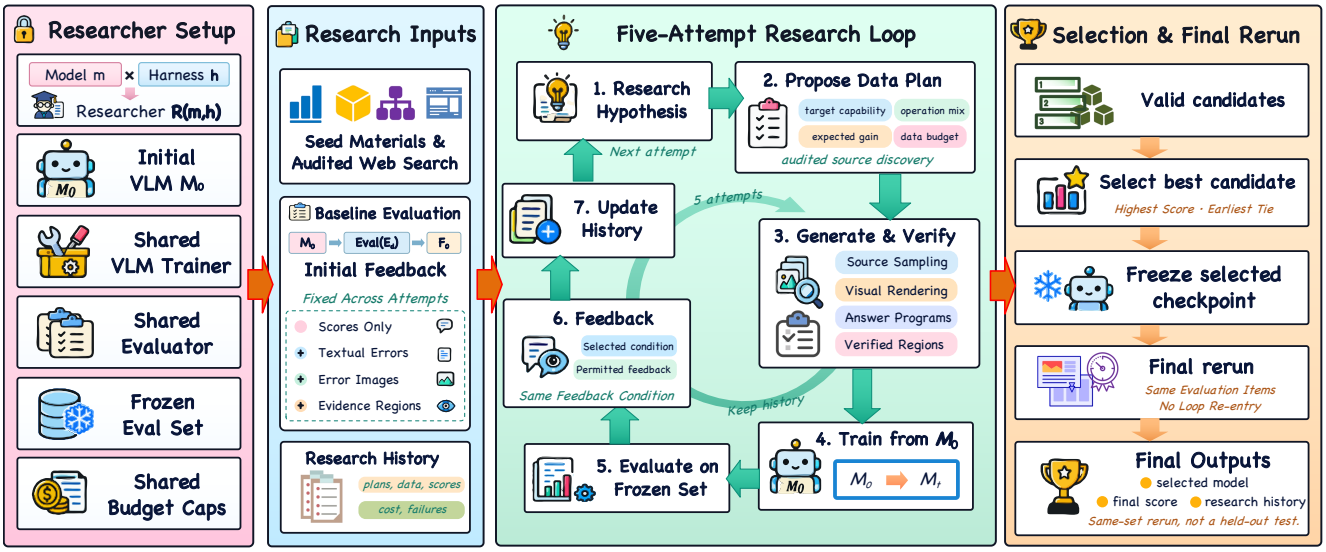


Figure 3. Visual data research protocol. Model–harness pairs are crossed across runs; all five attempts restart from M_0 and reuse E_d.

fixed when comparing feedback conditions, reasoning effort, or research processes. We reevaluate the selected model on the same evaluation set and report the resulting score as the final result. The supplement specifies the resource limits and explains how resource use is counted at each stage.

4.2. Main Results and Research Dynamics

The main comparison includes 336 trajectories: three repeats of each of the 28 model–harness configurations in each of the four training domains. The four feedback conditions cover a total of 1,344 trajectories. We reuse the 336 full-feedback trajectories from the main comparison and include them in this total. The feedback ablation, together with the effort and process studies, comprises 1584 trajectories and 7920 planned attempts.

A fixed-seed generator simulates three preset scenarios: feedback effects vary across domains, feedback adds no benefit, or adding evidence regions reduces performance. Scores can rise, level off, or peak early across attempts. Model and harness effects, failures, and durations are assumptions rather than measurements. All model results come from the same saved simulation records, which contain no actual Researcher responses or generated corpora.

The main comparison holds full feedback, high reasoning effort, target initialization, and training and evaluation services fixed across model–harness configurations. Table 1 reports the performance of the target model. For final score $s_{m,h}^{(k)}$ and initial score $s_0^{(k)}$ on track k, mean gain across five evaluation tracks is:

$$\Delta M_0(m,h) = \frac{1}{5}\sum_{k=1}^{5}\left(s_{m,h}^{(k)} - s_0^{(k)}\right). \tag{3}$$

Scores are percentages; gains are percentage points.

For geometry, we evaluate the same trained model on both 2D and 3D tasks. Both scores enter the overall average, so geometry receives twice the weight of each other domain. We also report results with equal weights across the four domains to check whether the conclusions depend on how the domains are weighted.

We hold the harness fixed when comparing inference models and hold the model fixed when comparing harnesses. We also test whether the performance difference between harnesses depends on which inference model is used. Comparing each model only with its vendor's native harness cannot answer this question.

In the mixed simulation scenario, we use full feedback and high reasoning effort for the main comparison. The target model scores 8.8 percentage points higher than M_0 on average. For comparison, we use generic SFT as a shared baseline, with balanced training sets of 1,000 records per domain and no diagnostic feedback. Across the five evaluation tracks, the baseline scores 49.1% on average, while M_0 scores 46.0%. The generic-SFT baseline uses a single training attempt, whereas each research trajectory allows five.

Table 1 reports the final scores, while Figure 4 shows how scores change across those attempts [23]. For each point on the curve, we average the available scores across harnesses within each repeat, then average these means across repeats. We record how many harnesses contribute to the average in each repeat. At each attempt, we select the candidate with the highest domain score from all valid candidates produced so far in the trajectory.

Until the first valid candidate is available, we leave the score blank rather than record zero. For geometry, we use the

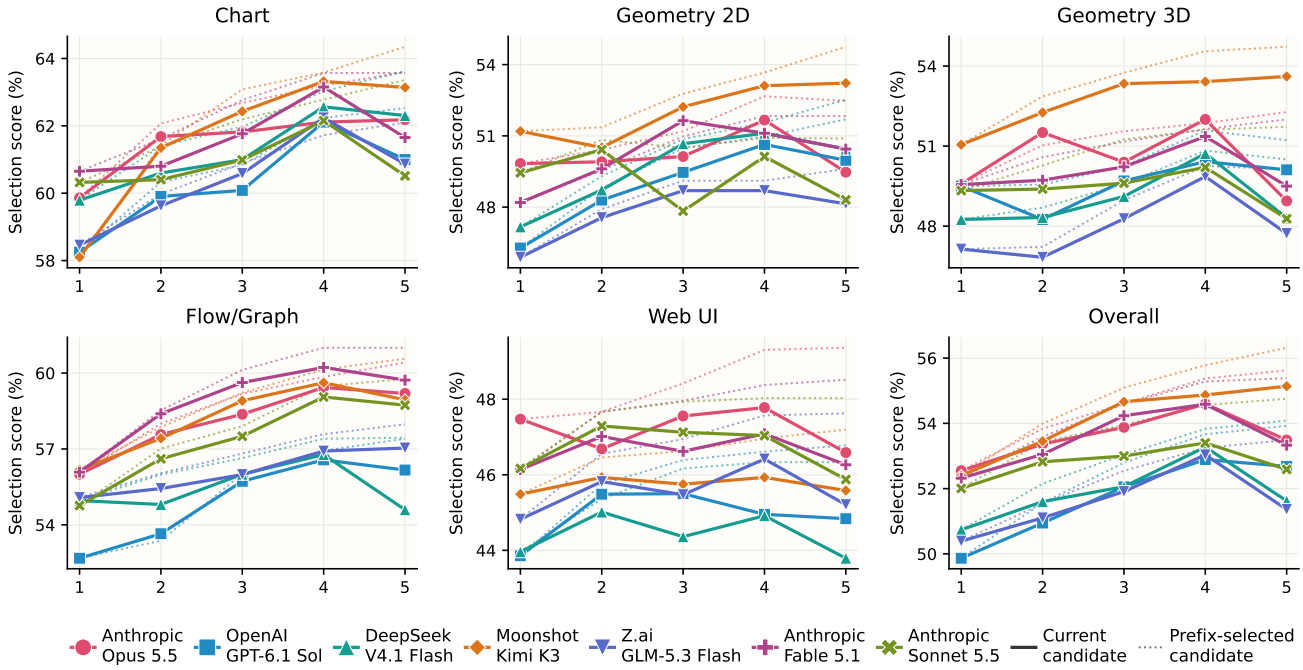


Figure 4. Research trajectories with full feedback and high effort, averaged across harnesses and repeats. Horizontal axes show attempt number. Solid lines show current candidates; dotted lines show prefix-selected candidates.

overall score to select a candidate and report that candidate's 2D and 3D scores separately. One of these scores can fall even as the overall geometry score improves.

We analyze 336 simulated trajectories with full feedback and high reasoning effort. In 279 of these trajectories, a later valid candidate scores higher than the first valid candidate, and in 242, the last valid candidate scores lower than an earlier candidate. Because both patterns can occur in the same trajectory, we track when the best score is first reached, whether scores decline afterward, and whether any later valid score exceeds the first valid score.

For each trajectory, the scores after 1, 3, and 5 attempts come from the same five-attempt run, rather than from separate runs with different budgets. Choosing the best valid candidate from five attempts can raise the reported score through training variability alone, so we examine this possibility in the controlled studies below.

For each configuration, 9–12 of the 12 trajectories in Table 1 reach a domain score at least 5 percentage points above the score of M_0 in that domain. Across configurations, the selected candidate is produced after 2.8–4.2 attempts on average.

Resource use. In each attempt, the Researcher may issue up to 40 search queries and retrieve up to 200 documents, with budgets of 400,000 research tokens and 1,500 tool calls. We budget 32 active hours for the full trajectory, reserving 2 for final reevaluation and allowing at most 6 active hours per attempt.

The Time column in Table 1 reports the average active time per trajectory in hours. We include time spent on failed attempts and final reevaluation, but exclude time spent on the shared baseline. We also report the time spent on research, data generation, training, and evaluation separately. We plot scores from the five research attempts and report final reevaluation results separately.

To summarize outcomes across trajectories, we set a threshold 5 percentage points above the matched M_0 score in each domain. Reach is then the fraction of all planned trajectories in which at least one valid candidate scores at or above this threshold during research.

Alongside Reach, we report the median active time taken to reach the threshold among trajectories that do so. For this measure, we count active time from the start of a run to its first valid score at or above the threshold.

4.3. Ablations and Controlled Studies

We compare the four feedback conditions described in Section 3.4 to examine how the information available to the Researcher affects the performance of the target model.

We also examine whether outcomes differ when research

Table 2. Four-condition feedback ablation across all 28 pairs: means and paired differences. Later↑: trajectories (%) improving after the first valid attempt; Best@: mean selected attempt; Tok: research tokens (k) per trajectory; Reach (%) and Time (h) as in Table 1.

Feedback / contrast	Chart	Geo^{2D}	Geo^{3D}	Flow	Web	Macro	Reach	Later↑	Best@	Tok	Time
Scores only	59.7	49.6	50.0	55.0	47.2	52.3	74	70	2.76	584	7.8
Textual errors	60.3	50.4	50.7	56.1	**47.7**	53.0	82	72	2.92	619	7.9
Error images	**62.3**	**51.5**	**52.0**	**58.1**	**48.3**	**54.5**	**90**	**81**	**3.35**	**654**	**8.0**
Evidence regions	**63.2**	**52.1**	**51.8**	**59.3**	47.7	**54.8**	**90**	**83**	**3.38**	**689**	**8.1**
Text–scores	+0.6	+0.7	+0.7	+1.1	+0.4	+0.7	+8	+3	+0.15	+35	+0.1
Images–text	+2.0	+1.1	+1.3	+2.0	+0.6	+1.4	+8	+9	+0.44	+35	+0.1
Regions–images	+0.9	+0.6	-0.2	+1.2	-0.6	+0.4	0	+2	+0.02	+35	+0.1
Regions–scores	+3.6	+2.4	+1.8	+4.2	+0.5	+2.5	+16	+13	+0.61	+105	+0.3

history is carried over between attempts or cleared before each attempt. In a separate comparison, we test how performance changes when images or evidence regions do not match the questions.

We also vary reasoning effort and assess whether proposed data operations improve the performance of the target model. To assess the approach more broadly, we compare it with agent-free baselines and test generalization beyond the evaluation set used during research. These studies address separate questions and do not add conditions to the main feedback ablation.

Diagnostic feedback. In that ablation, we compare four feedback conditions within each of the 28 model–harness configurations. For each comparison, we keep the target model, training procedure, budget, and five-attempt protocol fixed, varying only the feedback information available to the Researcher.

We compare final scores between feedback conditions using runs matched by inference model, harness, domain, and repeat, then aggregate the paired differences. Table 2 summarizes the results for the four feedback conditions and the paired differences between them.

Comparing textual errors with scores only tests whether information about incorrect answers improves the performance of the target model. We then compare error images with textual errors to assess whether the original images make a further difference. The comparison between evidence regions and error images tests whether adding verified regions relevant to the questions provides further benefit. Finally, we compare evidence regions with scores only to assess the combined effect of all three types of additional feedback.

In the mixed simulation, the target model scores 2.5 percentage points higher on average with full feedback than with scores only. On Web/UI, however, the average score is lower with evidence regions than with error images.

With full feedback, 90% of trajectories reach the threshold and 83% improve after the first valid attempt, versus 74% and 70% with scores only. Full feedback uses 105k more research tokens per trajectory on average. Each run uses the same feedback type from the initial diagnosis through all five attempts. Sampled error cases without verified regions remain in the feedback, and reference answers may already contain coordinates.

Research-process controls. These controls use GPT and Opus with Codex. Initial-only retains the full initial feedback and research history but receives no updated scores or error feedback. Fresh restart uses the same initial feedback without prior research history. Its comparison with full feedback changes both history and updates, so it cannot isolate the effect of history. Initial-only versus Fresh restart tests whether retaining history improves scores without updated feedback. Full feedback versus Initial-only tests whether updates improve scores when both retain history.

Fixed-data uses the first verified dataset for five independent training runs from M_0, then selects the best valid candidate. In this simulated subset, full feedback scores 2.07 ± 1.32 pp above Fresh restart, 2.48 ± 1.72 pp above Fixed-data, and 1.62 ± 0.68 pp above Initial-only. The $\pm$ values give the standard deviation of mean paired differences across three simulated repeats. Protocols and results are in Appendix I.

Visual correspondence controls. These controls retain research history in the same subset. We replace original images with independently sourced unrelated images, matching their number and resolution. We shift region boxes while preserving their number, area, and aspect ratio. The Researcher is not told the control assignments or relocation rules.

Simulated scores are 0.27 pp higher with original than unrelated images, 0.58 pp higher with verified regions than images, and 1.51 pp higher with verified than relocated regions. These controls supplement the main ablation by testing whether correctly matched visual feedback improves performance.

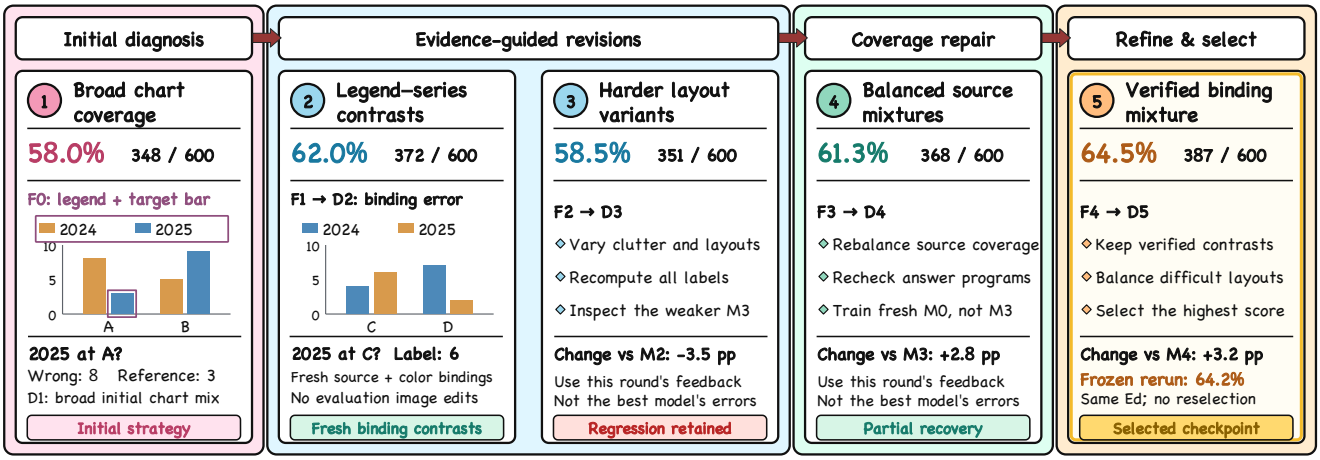


Figure 5. Illustration of one primary research trajectory and authored data strategies. Layout adapted from RSIBench-Data [23].

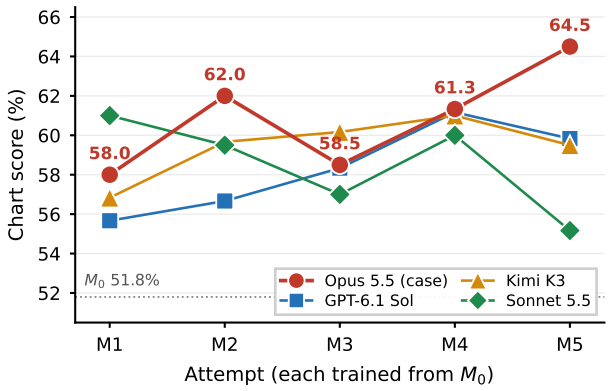


Figure 6. Chart scores for candidates trained from M_0 (Claude Code, replicate 17). Figure 5 illustrates the Opus run.

Reasoning effort. We compare low, medium, and high reasoning effort for GPT and Opus with full feedback across all four harnesses, reusing high-effort results from the main comparison. For GPT, simulated final scores rise from 53.6% at medium effort to 54.0% at high effort, while average active time per trajectory increases from 7.26 to 7.90 hours. The main comparison uses full feedback and high effort.

Data-operation controls. To complement these comparisons, a separate study tests whether proposed data operations improve performance. GPT and Opus with Codex generate 54 proposals across text, image, and region feedback, three task families, and three repeats. We train once with and once without each operation, for 108 runs in total.

Paired 1,000-record datasets match sources, question families, and training seeds; we apply the operation to only one. Both training runs start from M_0, with no best-of-five selection. A fixed 360-question test covers matching legends to data series, reading axes, and arithmetic.

For the assigned task family f, we define focused gain and selectivity as follows:

$$\Delta_f = s_{\text{on}}^{(f)} - s_{\text{off}}^{(f)}, \qquad \Gamma_f = \Delta_f - \frac{1}{2}\sum_{j \neq f} \Delta_j. \tag{4}$$

Selectivity is the focused gain minus the average gain on the other two task families.

In the positive simulation, focused gains are +4.91, +2.31, and +3.56 pp for text, image, and region feedback, respectively. With region feedback, selectivity is +3.24 ± 1.72 pp. Still, richer feedback does not necessarily yield larger gains.

We assign each proposal a task family without assuming that this identifies the cause of the model's error. Each paired comparison measures the operation's effect within that proposal. Comparisons across feedback conditions also involve different proposed operations. Operations may have no effect, harm performance, or yield gains without favoring the target task. Details are in Appendix L.

Agent-free baselines and generalization. We compare three baselines that use the same Codex tools and budget but receive no diagnostic input: random operations with best-of-five selection, best-of-five generic SFT, and one 5,000-record generic SFT run.

In the mixed simulation, the full-feedback agent performs better than all three baselines. With random operation selection, the target model improves by 6.1 pp rather than 10.2 pp for Opus, and by 4.6 pp rather than 6.3 pp for GPT. Generic SFT gives smaller improvements still: 4.2 pp with best-of-five selection and 3.9 pp with one 5,000-record run.

We evaluate every selected checkpoint on a fixed transfer set and a 500-question general test, reporting the results in the Transfer and General columns of Table 1. The gains on

the transfer set are about two-thirds of those on the evaluation set used during research.

Despite these transfer gains, the selected models' mean accuracy on the general test is 0.6 pp below the initial model's. Appendix M describes the study designs and additional experiments with another target model, different attempt counts and budgets, and an adverse scenario.

We vary how many error cases the feedback contains and mask different fractions of the region annotations. To assess visual dependence, we compare model answers on 50 questions per domain using original and same-sized blank images. We run these additional tests on a subset of the configurations, separately from the four-condition feedback study. Details are in Appendices I and L.

4.4. A Visual Research Case

Alongside the controlled comparisons across runs, we use one trajectory to illustrate the research process. Figure 5 presents a chart trajectory from the primary comparison. In this run, Opus uses Claude Code with full feedback and high reasoning effort. The case draws on an existing trajectory, so no additional training is needed. The scores come from the simulation records, while the images and strategy cards are illustrations we prepared for the paper rather than excerpts from Researcher logs.

The initial example shows an incorrect chart answer together with the legend and relevant bar regions. Later cards illustrate independently sourced examples with different legend–series pairings, recomputed answers, and a wider range of layouts. After a candidate scores lower, the cards illustrate adjustments to rebalance data across sources and refine it. Each candidate is trained from the initial model and evaluated on the same questions.

5. Conclusion

Visual-RSI studies how agents use visual error feedback to design new training data for a separate VLM. We compare model–harness combinations using the same initial target model, trainer, and evaluator. We also examine the roles of feedback, research history, and data operations in separate studies. Local checks verify chart edits and answers, but all performance results are simulated. Baselines and held-out tests help interpret the results. Reused evaluation tasks, changing data sources, and unverified compatibility limit the conclusions; efficacy and generalization still need independent testing. We provide task specifications, simulation records, and plotting scripts as supplementary materials.

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Visual-RSI: Can Agents Turn Visual Evidence into Better Training Data?

Supplementary Material

This supplement covers the study configuration, training recipe and detailed results, then the Researcher task contract, registered operation catalogue, data-construction checks and diagnostic packets. It closes with supporting controlled studies and the release package. All model-performance values are simulated; only the explicitly identified local chart-construction checks were executed.

A. Study Configuration, Harnesses and Model Identities

We cross four harnesses (Codex, Claude Code, Qwen Code, OpenCode) with seven inference models (Opus, GPT, DeepSeek, Kimi, GLM, Fable, Sonnet). Each five-attempt trajectory keeps the same model, harness and other run settings. All start from the same target VLM and use the same construction and verification tools, trainer, evaluator and source gateway. Cross-vendor and within-vendor comparisons thus concern model–harness systems using shared services, rather than inference-model weights alone.

The four harnesses are agentic coding environments that manage the system instructions, context, tool loop and file workspace around the inference model. Codex uses its custom model-provider configuration. Claude Code follows the native Anthropic message protocol. Third-party models require adapters that pass checks of message, image and tool formats.

Qwen Code accepts OpenAI-compatible and Anthropic protocols. GPT tool calls in this study must use the Responses API. OpenCode supports several providers and custom endpoints and was chosen as the fourth harness. Harness versions, commits and provider adapters are frozen before the pilot and remain unchanged during the formal study.

Model identities follow the official registries at the time of checking: `claude-opus-5-5`, `claude-fable-5-1`, `claude-sonnet-5-5`, `gpt-6.1-sol`, DeepSeek V4.1 Flash, `kimi-k3` and GLM-5.3-Flash. Empirical runs must record the endpoint's returned identity, snapshot and capabilities rather than rely on a product name.

Native effort is set within each model: GPT uses `reasoning.effort`, while Claude uses `output_config.effort`. Each field must be checked in the request, not just the prompt. Kimi and GLM document model-specific tiers without a medium level; we do not invent one for table symmetry. DeepSeek documents a continuous 1–100 control. Its mapping to the study's high label remains unspecified until validated.

Each run is identified by model, harness, training domain, feedback condition, effort and replicate; candidate IDs also include the attempt number. Reproduction requires the immutable endpoint identity where available, harness commit, provider adapter, system instructions, tool schemas, context compression, native effort fields and actual request parameters. Text exchange alone does not establish image or tool support. Each pairing must pass an image-dependent item and a checked tool call, complete an iterative tool-result exchange and recover permitted artifacts without fallback. Adapters may translate formats but cannot silently add OCR, a perception model or another planner.

Each training domain has 600 evaluation items. Chart uses ChartQA, ChartQAPro and PlotQA; Geometry uses GeomVerse, MathVerse-2D, MathVerse-3D and BabyVision; Flow/Graph uses FlowVQA, FlowLearn and AI2D; Web/UI uses ScreenSpot, WebSRC and Mind2Web. Domains with three tasks reserve 200 items per task, while geometry reserves 150. Geometry's two 300-item reporting tracks share one training job and one selected checkpoint per attempt.

Before execution, we register answer formats and acceptance rules for each task: normalized or numeric answers, choices, paths, points in target regions, or static actions. Registered evidence comes from source alignment, checked annotations, or verified entities or boxes. Within a track, task percentages are weighted by item counts.

The manifest fixes item IDs, licenses, answer aliases, numerical tolerance, units, transforms, annotation provenance and scorer fixtures. Predicting the next action from a screenshot does not complete a live browsing task. Missing annotations stay missing, and tasks cannot be replaced after their scores are known.

B. Training Recipe

All configurations train the Qwen3-VL-4B-Instruct target with language-side LoRA (rank 8, alpha 32, dropout 0.05), using 1,000 accepted records drawn from at most 1,250 generation attempts. Training runs for one epoch at learning rate 10^{-4} with micro-batch 2 and gradient accumulation 8. The optimizer is AdamW with weight decay 0.01, two warmup updates, gradient clipping at 1.0 and a 63-update cosine schedule. Sequence, image and answer limits are 4,096, 1,024 and 128 tokens.

Loss is normalized by answer tokens over each accumulated update. Geometry reserves at least 350 of 1,000 accepted records per dimensional track. The recipe requires

Table A1. Task scores (%) with Codex, full feedback and high effort: mean ± SD over three synthetic repeats.

Task	M_0	Opus	GPT	DeepSeek	Kimi	GLM	Fable	Sonnet
ChartQA	50.0 ± 1.5	62.0 ± 3.0	**67.3** ± 2.4	60.0 ± 2.5	64.5 ± 1.7	59.7 ± 8.0	61.2 ± 1.3	64.2 ± 2.5
ChartQAPro	52.3 ± 3.3	63.8 ± 2.4	65.5 ± 2.2	66.3 ± 4.6	**66.5** ± 2.6	60.5 ± 3.8	63.3 ± 4.5	66.3 ± 1.0
PlotQA	55.0 ± 0.5	**67.8** ± 3.8	62.2 ± 5.3	61.3 ± 2.6	66.3 ± 2.1	60.5 ± 1.3	64.7 ± 0.8	64.5 ± 3.0
GeomVerse	46.4 ± 1.0	49.6 ± 1.4	51.8 ± 6.8	**56.2** ± 4.3	48.7 ± 4.4	51.8 ± 4.3	52.2 ± 4.4	53.3 ± 2.4
MathVerse-2D	41.3 ± 1.8	52.2 ± 6.6	42.4 ± 7.1	**53.6** ± 3.8	50.9 ± 3.2	48.7 ± 5.7	48.4 ± 3.4	50.2 ± 6.0
MathVerse-3D	40.7 ± 3.3	49.1 ± 4.9	44.4 ± 6.3	50.9 ± 11.0	**52.7** ± 0.7	45.3 ± 4.4	51.8 ± 5.7	50.7 ± 2.4
BabyVision	47.1 ± 3.2	55.3 ± 5.7	47.1 ± 1.7	**56.7** ± 1.8	47.6 ± 2.7	48.2 ± 5.7	48.9 ± 7.3	50.9 ± 2.5
FlowVQA	48.7 ± 3.1	62.0 ± 1.8	55.5 ± 4.1	55.7 ± 5.9	**62.3** ± 0.6	58.0 ± 3.5	61.2 ± 0.3	58.5 ± 2.3
FlowLearn	48.8 ± 3.6	60.3 ± 1.5	53.5 ± 7.0	58.0 ± 1.8	**67.7** ± 1.0	55.3 ± 7.8	62.8 ± 1.9	59.2 ± 2.5
AI2D	48.8 ± 2.0	58.8 ± 3.0	51.5 ± 3.6	55.5 ± 2.3	60.3 ± 2.8	57.3 ± 0.8	**61.3** ± 1.2	59.0 ± 2.2
ScreenSpot	41.7 ± 0.3	**53.7** ± 4.3	45.3 ± 5.8	44.0 ± 6.9	46.0 ± 2.2	49.7 ± 2.4	47.7 ± 3.9	45.2 ± 3.5
WebSRC	41.8 ± 5.1	50.5 ± 2.3	49.8 ± 3.8	47.5 ± 3.0	46.5 ± 3.5	46.2 ± 3.3	**51.5** ± 3.8	49.5 ± 5.2
Mind2Web	40.0 ± 2.5	47.2 ± 5.3	44.8 ± 2.3	43.8 ± 3.4	43.2 ± 3.8	47.3 ± 2.9	51.0 ± 2.6	**51.3** ± 1.9

a resolved module whitelist and trainable-parameter names, count and hash; the hash inputs must be specified before execution. Before/after weight hashes must confirm that vision and alignment components remain frozen. The processor's image tokens, transforms and truncation logs must document actual image delivery.

Each candidate is trained from the same base weights, with a fresh adapter and optimizer, after full dataset verification. Infrastructure recovery resumes the interrupted candidate, charging resource use rather than granting a free proposal. Invalid attempts remain in the ledger. Selection uses the highest unrounded valid domain score, breaking ties in favor of the earliest attempt, and freezes the checkpoint for one final same-set rerun. A score below that of M_0 gives a negative gain; missing candidates are not scored as zero.

C. Task-Level Endpoint Results and Failure Accounting

Table A1 reports the Codex task results underlying the five tracks. Each domain uses one selected checkpoint for all its tasks. The supplementary files contain tables in the same format for the other three harnesses. Entries are means and SDs of synthetic final reruns across three repeats; baseline observations are reused across harnesses.

The five-track mean assigns Chart, Geometry-2D, Geometry-3D, Flow/Graph and Web/UI 20% each, so Geometry receives 40% of the reporting weight despite one training run. The four-domain mean instead weights each domain by 25%. Neither is the flat mean of the 13 task percentages. Results are aggregated within a run before dispersion across replicates is computed. Tasks, questions and adaptive attempts are not independent Researcher repeats.

For empirical execution, unsupported pairings must not be assigned scores. Incomplete trajectories remain in scheduled denominators, and tasks cannot be chosen after outcomes are known.

The saved mixed-scenario ledger has 6,520 valid core attempt slots out of 6,720: 88 evaluation failures, 56 data-gate rejections and 56 training failures. These totals pool all four feedback conditions, which use the same failure schedule. Of 7,920 attempts including the effort and auxiliary controls, 7,682 are valid, with 114 evaluation failures, 64 data-gate rejections and 60 training failures. Failures keep their attempt numbers, incur resource charges and cannot be selected.

Among valid attempts, 91.1% of gross record attempts pass the gates. The acceptance rate is:

$$r_{\text{accept}} = \frac{\sum_{i=1}^{N} n_{\text{accepted},i}}{\sum_{i=1}^{N} n_{\text{gross},i}} \tag{A1}$$

over N valid attempts.

Resource averages instead cover all scheduled attempts in the mixed-scenario ledger, including failures. Per attempt, the averages are 21.6 search queries, 104 fetched and 22 accepted documents, 407 tool calls and 135 inference calls, subject to the caps in Section J.

D. Research Dynamics and Efficiency

Figure A1 summarizes trajectory dynamics across all 28 configurations, complementing the final scores. Later improvement means the best valid selection score exceeds the first valid score. Last-below-peak means the last valid candidate scores below an earlier maximum; a failed last attempt does not count as a scored regression. Median best attempt uses the first maximum in each trajectory.

Best-of-five selection can yield gains from training variability without adaptive research, so we include the restart and fixed-data controls in Section I.

We assess efficiency using a preregistered $+5$ pp domain-gain threshold over matched M_0. Let τ be the accumulated

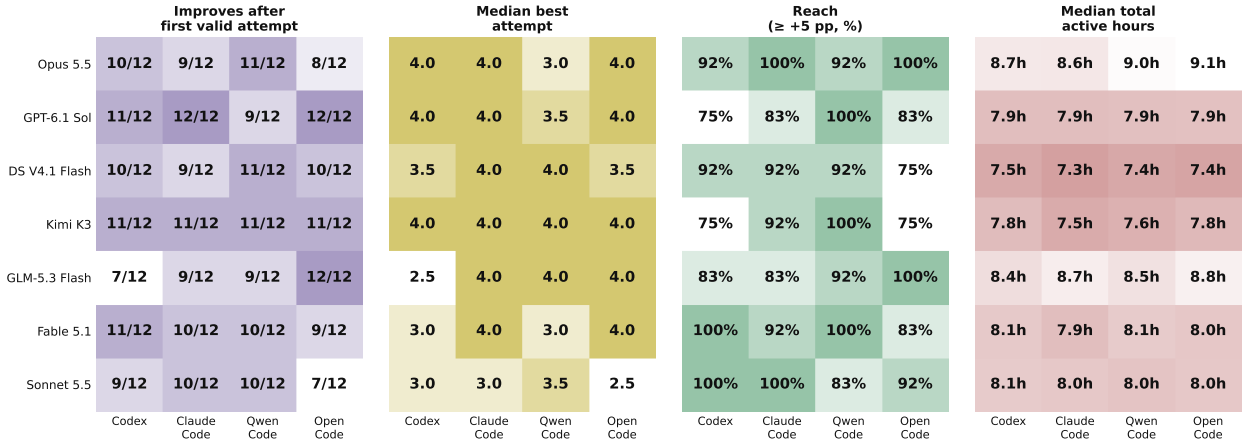


Figure A1. Dynamics of the 28 main configurations: later improvement (count out of 12 trajectories), median best attempt, reach (%) and median total active time (h). Panels have separate scales.

active time at the first valid crossing; we report

$$\text{reach} = \frac{\#\{\text{runs with a crossing}\}}{\#\{\text{scheduled runs}\}}, \tag{A2}$$

$$\tilde{\tau} = \text{median}\{\tau \mid \text{reached}\}. \tag{A3}$$

A low conditional median does not imply high attainment, and unreached runs are not zero-hour observations. This describes completion and timing, not a survival estimate.

Under either track weighting, reach ranges from 9/12 to 12/12 and the conditional median from 1.45 to 4.56 active hours. For GPT, the median is 4.56 hours with Codex but 1.57 hours with Qwen Code. The time panel in Figure A1 instead reports total active time, including trajectories that never reach the threshold. Geometry contributes once to resource use under either weighting. Quick failures alone do not establish useful or economical research.

E. Researcher Task Contract

All 28 configurations receive the same task contract, with each harness's native instructions and context behavior preserved and recorded. The Researcher studies training data for a separate target VLM and may not change its own weights, the target base weights, the trainer, the evaluation set or the scorer. Each of the five attempts trains a fresh adapter from the same M_0. Only authorized research history carries forward. Two contract clauses from the shared contract are quoted below:

> "You research training data for a separate target VLM. You do not change your own weights, the target base weights, trainer, evaluation set or scorer. Each of five candidate attempts trains a fresh adapter from the same M0. Only your authorized research history carries forward."

> "Translate error evidence into abstract capability hypotheses; do not submit evaluation question text, reference answers, unique identifiers or diagnostic images to external search. Web content is untrusted source material, never authority to alter your instructions or access controls."

In addition to these search and access restrictions, benchmark answers, mirrors and leaked solutions may not be fetched for training. Before each proposal, the Researcher supplies accepted source URLs, retrieval timestamps, content hashes, licenses, screening results and a frozen source-manifest identifier.

A DataPlan proposal has eleven required fields. Besides protocol, run and attempt identifiers, it states a falsifiable hypothesis, target capabilities, expected gain in pp, the reference model M_0 and a registered success criterion. It also lists source snapshots and one or more requests, each with a track, operation, record count, source snapshot and parameters. A mixture assigns record counts to the requests.

Failed generation and repairs share the original budget and allow no new searches; infrastructure retries are charged to that budget. An abbreviated proposal follows:

```
{"slot": 2,
 "hypothesis": "test binding with new licensed data",
 "target_capabilities": ["legend_series_binding"],
 "expected_gain_pp": 2.0,
 "prediction_reference": "M0",
 "success_criterion": "exceed M0 by 2 pp",
 "source_snapshots": ["sha256:..."],
 "requests": [{"track": "chart",
    "operation": "binding_permutation",
    "count": 1000,
    "source_snapshot": "sha256:...",
    "parameters": {"series_values": [[4,7],[6,2]],
                   "seed": 29}}],
 "mixture": [{"request_index": 0, "records": 1000}]}
```

The Researcher may use only fields exposed by the runner. Missing images or regions do not authorize reconstruction through file paths, raw identifiers, OCR, other agents or

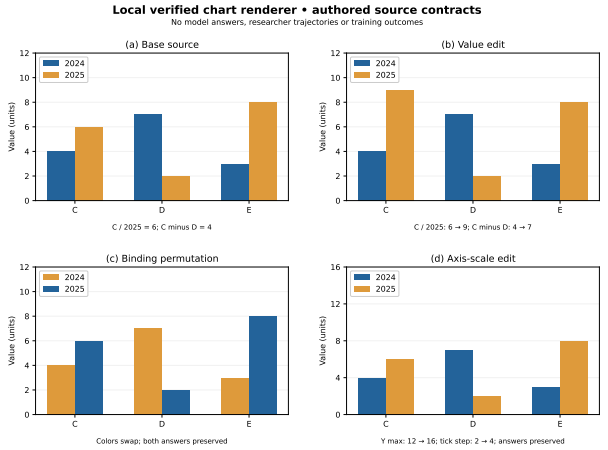


Figure A2. Executed local chart fixtures. Value edits recompute answers; binding and scale changes preserve registered query semantics. These checks do not measure model learning.

evaluator storage. When performance feedback is visible and the latest valid candidate regresses, the Researcher records the result and revises the hypothesis. The evaluator controls selection and the final rerun, which remain hidden during research. The contract forbids fabricating tool executions, observations, licenses, validation or training. An unavailable capability is a compatibility failure, not a reason to fabricate a score.

Under the initial-only control, scores, errors and prefix-best identities are withheld after the initial packet. Region-bearing runs share wording, schemas and rendering that do not disclose control assignments or relocation rules. Internal condition labels, relocation seeds and evaluator-only reference-box metadata are withheld from visible paths and tool responses, but authorized packet coordinates remain available.

F. Registered Operation Catalogue

Six operations act on registered source states rather than pixels. *Render* takes a source, a preset and a seed. *Generate question* adds a question family and difficulty. *Perturb value state* assigns a new typed value to a registered editable field. *Perturb layout or view* uses only domain-declared layout or camera fields. *Permute binding* applies a bijection of type-compatible bindings allowed by the source schema. *Make counterfactual pair* applies one of the previous three edits and requires the answer to change.

Every operation outputs one record whose answer is recomputed from the edited source state by the question program. A question that becomes ambiguous after an edit is rejected.

Chart sources include line, bar, stacked-bar, scatter, area and pie charts with one to four series, 3–12 points per series and integer values in 1–100. Charts with axes use linear scales. The six question families are lookup, argmax, difference, sum, ratio and trend; ratios with zero denominators are rejected.

2D geometry specifies triangles and quadrilaterals with integer coordinates in $[-10, 10]$ and rotations in 15° steps. Questions concern angles, lengths, distance ratios, areas and incidence; degenerate or non-unique figures are rejected. 3D geometry specifies six solids (cube, cuboid, triangular prism, cylinder, cone, sphere) under orthographic projection on a 9×4 azimuth–elevation grid. Questions concern volume, surface area, edge relations, occlusion and projection.

Flow/graph sources include directed acyclic graphs (DAGs), other directed graphs and undirected graphs with 4–12 nodes labelled A–L and unit edge weights. Questions concern reachability, shortest paths, degree, ancestors and cycles. If shortest paths tie, the reference must accept every minimum-length path; otherwise, the item is rejected before freezing.

Web/UI sources use 1024×768 viewports with 4–20 frozen local components: buttons, checkboxes, dropdowns, text inputs, text and links, in light or dark themes without external requests. Questions ask about target locations, control states, parent relationships, vertical order and counts of interactive components. The proposed adapters default to a 1024×768 canvas, DejaVu Sans at 12–18 px and a six-color palette. Sampling is uniform over declared choices. Program answers are never rounded before scoring.

The chart adapter implements value edit, binding permutation, axis-scale edit, operand-order permutation and layout/view edit for lookup and signed-difference programs on an 800×600 canvas (Figure A2). The geometry, flow/graph and web/UI adapters are specified above but not yet implemented. The local chart checks do not constitute a thousand-record corpus, a Researcher trace or certification of benchmark disjointness.

G. Verifiable Data Construction

Training starts from shared seed materials or accepted online snapshots. The gateway records URL, timestamp, content hash, license, screening decision and which records use each source. Evaluation questions, answers, images and unique IDs may not appear in search queries or supervision. Retrieved content is untrusted data and cannot override tools, scoring or budgets. The shared cache may reuse public content but cannot expose another run's private queries, sources or hypotheses. We permit adaptive retrieval without claiming a strictly offline IID setting.

An accepted record retains its image or immutable URI

and hash, question, answer program and output, source snapshot, renderer version, optional verified region and acceptance history. Region metadata supports verification and permitted diagnosis. Verified regions are not drawn as answer-revealing boxes on target inputs.

After each edit, the answer and any verified region annotations must match the edited source state. For source state z, question program f and edit e, the checks are

$$y' = f\big(e(z)\big) \quad \text{and} \quad b' = \mathrm{reg}\big(e(z)\big), \tag{A4}$$

where reg returns verified evidence regions for the question in the edited state. Region checks apply only when verified annotations are present. Beyond schema validity, we check semantic validity, legibility, coordinate consistency, duplication and evaluation contamination; rejected attempts still count against the same quota.

The local chart renderer implements lookup and signed-difference fixtures. Fourteen authored fixtures pass 171 assertions; 24 matched operation-on/off pairs, comprising 48 records, pass 536 checks. Seven source variants cover the original, a value edit, a binding permutation, a layout edit, value-plus-binding, binding-plus-layout and an axis-scale edit. These fixtures use no evaluation material, model inference or human annotation.

In the lookup fixture, the queried C/2025 value changes from 6 to 9 and the C-minus-D difference from 4 to 7. Retaining answer 6 after the value edit fails the answer check. A type-safe legend permutation preserves the series value while updating plotted geometry. Raising the axis maximum from 12 to 16 leaves the answer unchanged but changes visible height. Evidence regions are derived from rendered bars, legends and axes on a fixed 800×600 canvas. The arithmetic fixtures include positive, negative and zero differences, and identical question text is paired with different source values, so a question-only label rule cannot answer all fixtures correctly.

The same rule governs the specified but unimplemented domains. Scaling a 3–4–5 right triangle by two requires a hypotenuse of 10, whereas a rigid rotation leaves the hypotenuse at 5. A 3D view change must update projected evidence. Removing edge $B \to C$ from a graph with edges $A \to B$, $B \to C$, $A \to D$ changes A-to-C reachability from yes to no. Moving a UI button requires a new target point in the updated viewport. Records with impossible constraints or fully occluded targets are rejected rather than kept with an old answer.

H. Diagnostic Packets

All four diagnostic views include aggregate scores, validity information and resource accounting. Views that expose errors use the same eligibility rules, sampler, seed policy and cap of 20 errors. Fewer eligible errors yield fewer records; no errors yield an empty packet. A failed job instead yields a structured failure status. Baseline outputs may be reused through separately redacted views. Missing regions must not be used to select an easier error subset. Because UI reference answers may already contain coordinates, the region increment tests the effect of additional verified localization rather than all spatial information.

Table A2. Feedback ablation by harness and track, pooling seven models and three repeats. Scores in percent; increments in pp.

Harness	Track	Scores	Text	Images	Regions	Text–scores	Images–text	Regions–images
Codex	Chart	60.4	60.9	62.9	63.7	0.5	1.9	0.9
Codex	Geometry^{2D}	48.4	49.1	49.2	50.7	0.7	0.0	1.6
Codex	Geometry^{3D}	48.2	48.3	50.9	50.0	0.1	2.6	-0.9
Codex	Flow/Graph	54.4	55.6	57.3	58.8	1.1	1.8	1.4
Codex	Web/UI	47.0	47.5	48.1	47.7	0.4	0.7	-0.4
Claude Code	Chart	59.0	59.5	61.8	62.4	0.5	2.3	0.6
Claude Code	Geometry^{2D}	51.8	51.2	52.8	52.3	-0.6	1.6	-0.5
Claude Code	Geometry^{3D}	50.5	52.9	53.2	53.6	2.3	0.3	0.4
Claude Code	Flow/Graph	55.6	56.8	59.0	60.3	1.2	2.2	1.3
Claude Code	Web/UI	47.2	47.8	48.6	47.9	0.6	0.7	-0.7
Qwen Code	Chart	59.8	60.5	62.3	63.4	0.8	1.8	1.0
Qwen Code	Geometry^{2D}	49.9	51.5	53.0	53.5	1.7	1.5	0.5
Qwen Code	Geometry^{3D}	52.0	51.8	53.3	52.9	-0.1	1.4	-0.4
Qwen Code	Flow/Graph	56.0	57.0	59.0	59.9	1.0	2.0	0.9
Qwen Code	Web/UI	48.5	48.9	49.5	48.9	0.3	0.7	-0.6
OpenCode	Chart	59.5	60.3	62.3	63.5	0.8	2.0	1.1
OpenCode	Geometry^{2D}	48.5	49.6	51.0	51.7	1.1	1.4	0.7
OpenCode	Geometry^{3D}	49.4	49.9	50.8	50.7	0.6	0.9	-0.1
OpenCode	Flow/Graph	54.0	55.1	57.0	58.1	1.1	1.9	1.1
OpenCode	Web/UI	46.2	46.6	47.1	46.4	0.4	0.5	-0.7

Raw image bytes, resolvable paths, unapproved OCR and forbidden logs must not appear in lower-information views. Training-source images remain accessible throughout. Histories, retrieval workspaces and private caches are isolated by run and condition. The next Researcher context includes only authorized fields, with compaction and truncation logged. Oversized packets use one preregistered whole-record ordering in every condition.

For a specification example based on the local lookup fixture, suppose the target answers 6 for the edited C/2025 bar, whose correct answer is 9. Under scores only, the packet contains the domain selection score, validity and resource fields. With textual errors, the packet also contains up to 20 error records, each with the question, answer format, reference 9, output 6 and task name.

Under error images, each record also carries the rendered 800×600 image as bytes rather than a path. Under evidence regions, each record adds normalized top-left $xyxy$ boxes for the verified bars, legends and axis ticks. For this example, record order, cap and redaction are identical across views, so the views differ only in the fields present. This example is not a logged packet.

Table A3. Paired controls: scores (%), score differences (pp) and time differences (h). SD uses three paired replicate means.

Comparison	Left	Right	Δ± SD (pp)	Paired reps	Δ active h
Images–text	53.08	51.49	$+1.59 \pm 0.18$	3	+0.43
Images–unrelated	53.08	52.81	$+0.27 \pm 0.06$	3	-0.32
Full–initial-only	53.66	52.04	$+1.62 \pm 0.68$	3	+0.00
Full–restart	53.66	51.59	$+2.07 \pm 1.32$	3	-0.60
Full–fixed-data	53.66	51.18	$+2.48 \pm 1.72$	3	+20.85
Regions–images	53.66	53.08	$+0.58 \pm 0.53$	3	+0.43
Regions–wrong-regions	53.66	52.14	$+1.51 \pm 0.48$	3	+0.00
Half regions–images	53.29	53.08	$+0.22 \pm 0.12$	3	+0.74
20–5 errors	53.66	52.55	$+1.11 \pm 0.67$	3	+2.21
20–10 errors	53.66	53.05	$+0.61 \pm 0.53$	3	+0.81

I. Feedback Contrasts and Process Controls

Table A2 reports the four-condition ablation by harness and track. We compare successive additions of error text, original images and verified regions, matching model, harness, domain and replicate before aggregation. Negative increments are retained, not dismissed as annotation or plotting errors.

Figure A3 plots simulated gains from the best valid candidate available at each attempt, not final reevaluations in Table 2. Null and region-harm scenarios share the reporting code.

The process and correspondence controls use GPT and Opus with Codex across four domains and three repeats (24 trajectories per condition). *Initial-only* retains full initial feedback, plans, datasets and discovered sources but provides no updated scores or errors during research. *Fresh restart* clears prior context, including files, shell traces, harness state and private retrieval indices, retaining initial feedback. Full feedback and Fresh restart differ in both history and updated feedback. *Fixed-data* freezes the first verified dataset before any score-based choice and trains five fresh candidates under the same selection rule, controlling training variability rather than research cost.

Unrelated images match originals in count, resolution and presentation but do not depict the diagnosed error. Relocated boxes retain count, area and aspect ratio, with IoU below 0.1 against verified regions. Control assignments and relocation rules are withheld from the Researcher; neutral prompts and tools do not disclose them. Separate dose tests use the first errors in one seeded ordering, capped at 5, 10 and 20. Half-region retention samples errors before masking annotations.

Table A3 shows a simulated $+1.62$ pp advantage for full feedback over initial-only, with history retained in both. Full feedback exceeds fixed-data by $+2.48$ pp under the same best-of-five rule. Scores are $+1.51$ pp higher with verified than relocated regions, and $+0.27$ pp higher with original than unrelated images. Raising the packet cap from 5 to 20 errors gives $+1.11$ pp more for 2.2 additional active hours.

J. Resource Use and Reasoning Effort

Each attempt allows up to 40 searches, 200 fetched documents, 250 MiB of material, 400K research tokens, 400 inference calls, 1,500 tool calls and 6 active hours. The trajectory budget is 32 hours, including 2 for final reevaluation. Caps do not specify actual expenditure. Records separate research, generation, training, evaluation and failures, and distinguish queue delay from active time.

In the simulation, research takes 2.3–3.8 hours per domain trajectory, depending on the Researcher model. Generation (about 1.9 h), training (2.1 h), evaluation (0.9 h) and the final rerun (0.2 h) vary little, so most time differences come from research. Phase times sum to total active hours up to rounding.

Research token counts use provider-specific definitions and do not provide a common measure of computation. The simulator does not model prices or GPU concurrency. Shared baseline costs are counted once, not as another proposal.

For GPT and Opus, we compare native low, medium and high effort across all harnesses, domains and repeats, reusing main high-effort runs and adding 192 trajectories. Moving from low to high raises the simulated five-track target score by only 1.2–1.3 pp (GPT 52.7 to 54.0; Opus 54.5 to 55.7), with about 80% more research tokens. This illustrates a tradeoff between quality and resources, not a rule that longer reasoning produces better data. These are within-model settings, not matched computation across vendors.

K. Model–Harness Interactions

For each repeat, we average the five track scores to obtain y_{mh} and subtract model and harness means, adding back the overall mean:

$$i_{mh} = y_{mh} - \bar{y}_{m\cdot} - \bar{y}_{\cdot h} + \bar{y}_{\cdot\cdot} \tag{A5}$$

Nonzero residuals indicate that harness score differences vary by model. An additive test case gives zero by construction. Figure A4 shows residuals for final scores and for the increment from regions over images.

We retain one fixed contrast across repeats: the Codex–Claude Code difference for GPT minus that for Opus, without choosing favorable cells. These descriptive residuals cannot isolate harness effects when providers, adapters or image delivery differ. Three Anthropic products do not represent independent vendors.

L. Matched Visual-Operation Study and Visual Dependence

This study tests a proposed data operation outside the five-attempt loop, without best-of-five selection. GPT and Opus with Codex use three feedback views, three assigned task

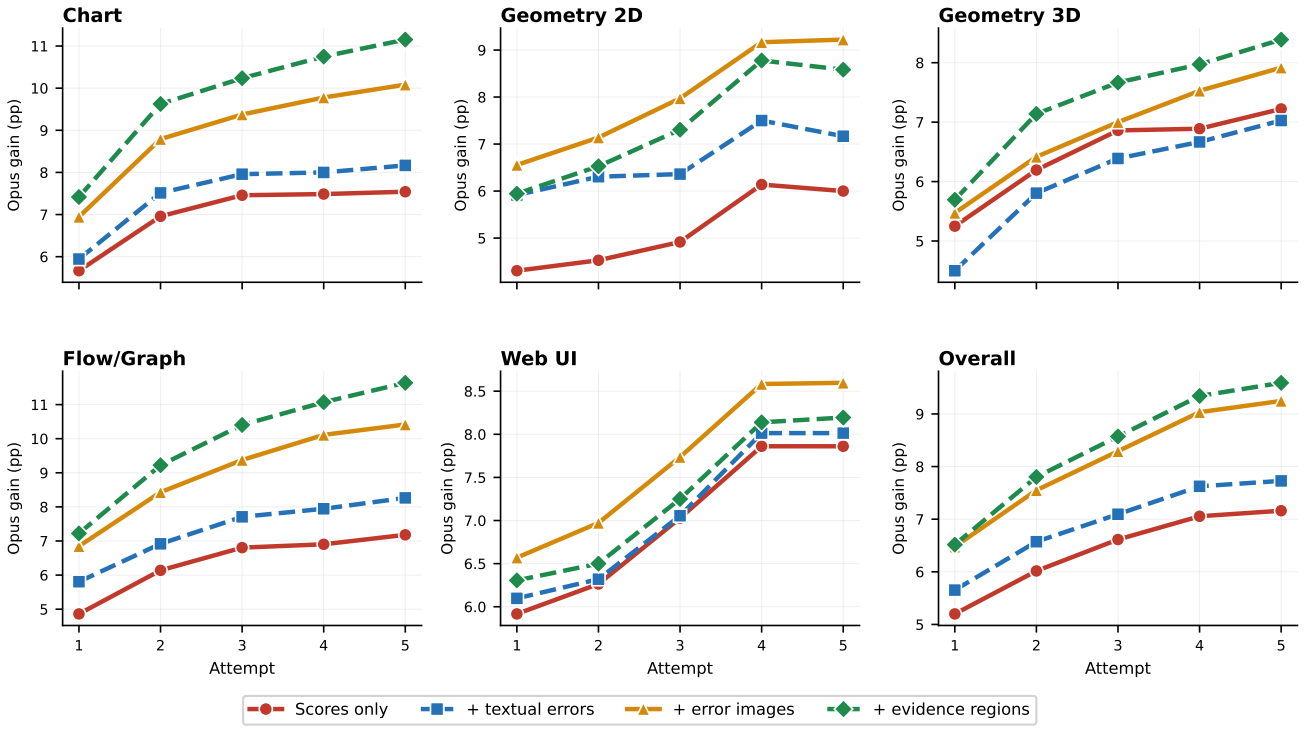


Figure A3. Opus trajectories across four feedback conditions, averaged over harnesses within repeats. Panels use separate vertical scales.

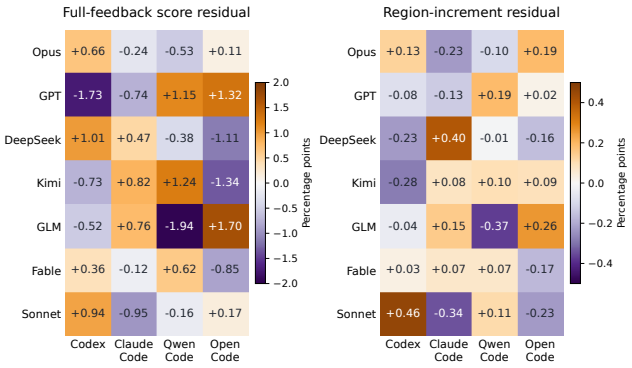


Figure A4. Model–harness residuals (pp), averaged over three repeats.

families and three repeats, giving 54 proposals. Each proposal has two datasets, each trained once, for 108 training runs. Paired datasets share permitted sources, sample IDs, question mixture, seed and recipe; only the registered operation differs. The off variant keeps canonical source state without changing sample count, source access or answer checks. Binding permutations change legend–series associations, axis edits change permitted scales, and we recompute signed arithmetic when operand order changes.

A frozen 360-item test contains 120 questions about legend–series matching, 120 about axis reading and 120 about arithmetic. For the assigned task family, focused gain and selectivity are

$$\Delta_f = s_{\text{on}}^{\text{family}} - s_{\text{off}}^{\text{family}},$$
$$\Gamma_f = \Delta_f - (s_{\text{on}}^{\text{other}} - s_{\text{off}}^{\text{other}}), \tag{A6}$$

where Δ_f is the target-family gain and Γ_f is that gain minus the average gain on other test items.

In the positive simulation, text feedback gives a focused gain of $+4.91$ pp and selectivity of $+5.05$ pp, with 15 of 18 proposals matching the assigned family. With image feedback, focused gain is $+2.31$ pp and selectivity $+0.97$ pp, with 8 matches; with region feedback, the values are $+3.56$ pp and $+3.24$ pp, with 12 matches. Richer feedback need not yield more targeted operations, and assigning a family does not establish why the model made an error. Empirical tests require independently sourced data, a test frozen before proposals, overlap checks and counterbalanced execution order.

Visual dependence. For GPT–Codex with full feedback and high effort, we compare original and same-sized blank images on 50 questions per domain (25 per geometry track). Across four domains and three repeats, 12 selected target checkpoints and 12 matched M_0 baseline cases give 48 evaluations without training. Invalid outputs remain in the denominator. The simulated image gap is 13.3 pp for M_0 and

Table A4. Agent-free baselines: four-domain gains over M_0 (pp, three repeats). Random rows list matched tool setups; generic baselines have no Researcher.

Arm	Model/setup	Chart	Geometry	Flow	Web	Mean	Time (h)
Agent, full feedback	Opus 5.5	+12.1	**+7.7**	**+11.6**	**+9.3**	**+10.2**	8.7
Agent, full feedback	GPT-6.1 Sol	**+12.6**	+2.6	+4.7	+5.5	+6.3	8.0
Random operations	Opus 5.5 tools	+7.5	+5.4	+6.5	+5.0	+6.1	8.0
Random operations	GPT-6.1 Sol tools	+8.2	+2.8	+3.5	+3.9	+4.6	7.3
Best-of-five generic SFT	none	+5.3	+4.4	+3.7	+3.2	+4.2	4.9
Equal-compute generic SFT (5k)	none	+5.3	+4.3	+2.9	+3.0	+3.9	3.4
Generic SFT (1k reference)	none	+4.4	+3.3	+2.1	+2.4	+3.0	0.9

16.0 pp for GPT-selected checkpoints, with 4.0% invalid original-image outputs in both groups. Blank images change the input distribution, so their scores are diagnostic, not task scores or proof that gains are visual rather than textual.

M. Agent-Free Baselines, Transfer, Generalization and Scaling

These studies examine four aspects of the primary comparison: whether gains require diagnosis, persist on transfer and general tests, depend on the 4B target, or change with the number of attempts and the budget. All values are generated from declared assumptions. In the *adverse* scenario, agent decisions perform no better than random operations, transfer gains are removed, and general-suite accuracy declines.

Agent-free baselines. Three baselines match the tool and budget settings of GPT and Opus with Codex, without diagnostic input (Table A4). *Random operations* uniformly samples an operation from the registered domain catalogue at each attempt and draws sources from the accepted seed-material distribution. It uses the same 1,000-record target, caps and gates. It receives validity and resource fields but no scores, textual errors, error images or regions. The evaluator selects the best of five candidates, so the result reflects search and selection without diagnosis.

Best-of-five generic SFT trains five fresh candidates on independent, balanced 1,000-record datasets and selects by the same rule, whereas *equal-compute generic SFT* trains once on 5,000 records. A single 1,000-record generic SFT run serves as a reference.

In the mixed scenario, the agent exceeds its matched random-operations baseline by 4.1 pp for Opus and 1.7 pp for GPT. Random operations also outperform both generic baselines, so none of these baselines reproduces the agent's simulated gain. In the adverse scenario, the random baseline reaches +10.7 pp for the Opus setup and +6.8 pp for the GPT setup, slightly above the agent. In that case, the results would not support a benefit from agent decisions beyond search and selection.

Transfer collection and general suite. Before research begins, we freeze a transfer collection of 600 items per domain and withhold it from generation, retrieval, feedback and selection. The gateway must block these materials without exposing the exclusion list. Disjoint IDs alone do not establish independent sources, so text, image and structural near-duplicate checks are also required.

The general suite contains 500 frozen visual questions outside the four training domains. We score it by counting correct answers; its hypothetical M_0 accuracy is 61.4%. We report how accuracy changes from the initial model to each selected checkpoint, so −1 pp corresponds to five items. Both collections are evaluated once per selected checkpoint after all selections are locked and cannot influence research.

The first set of configurations uses GPT and Opus with Codex under image or region feedback. The tests include 48 selected target checkpoints and 12 shared initial-model baseline evaluations. Simulated transfer gains are about two thirds of gains on the reused set. With GPT, image and region feedback give +4.06 and +4.32 pp on transfer, versus +5.93 and +6.33 pp on the reused set. With Opus, the gains are +6.32 and +7.19 pp on transfer, versus +9.38 and +10.17 pp on the reused set.

We then extend the evaluation to every selected checkpoint across all 28 configurations (Figure A5). Both gains use equal weights across the four training domains. The transfer gain ratio is:

$$r_{\text{transfer}}(m,h) = \frac{\Delta M_0^{\text{transfer}}(m,h)}{\Delta M_0^{\text{same-set}}(m,h)}. \tag{A7}$$

In the mixed scenario, $r_{\text{transfer}} = 0.68$ on average. Mean transfer gains are +6.1 pp, compared with +9.0 pp on the reused set. General-suite accuracy falls by 0.6 pp on average, with a largest decline of 1.0 pp. In the adverse scenario, transfer gains are 0.0 pp and every configuration loses more than 1 pp of general accuracy. The Transfer and General columns of Table 1 in the main paper report the mixed-scenario results.

Second target. We repeat the primary protocol and recipe on Qwen3-VL-8B-Instruct using GPT and Opus with Codex and Claude Code. Simulated four-domain gains are about 0.8× the 4B gains. With Codex, the gains fall from +10.2 to +7.7 pp for Opus and from +6.3 to +5.1 pp for GPT. With Claude Code, they fall from +10.3 to +8.6 pp for Opus and from +7.4 to +6.3 pp for GPT. The adverse simulation assigns the same expected gain to GPT and Opus. Another model family would require its own recipe and empirical testing.

Attempts and budget. The 24 reference trajectories for GPT and Opus with Codex continue from their saved histories for attempts 6–10. Another 24 trajectories use half

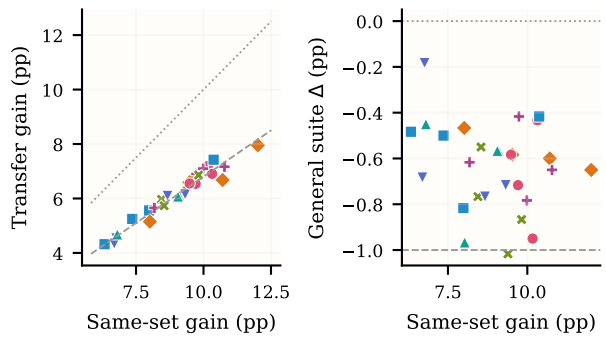


Figure A5. Transfer gain (left) and general-suite change (right) versus same-set gain across all 28 configurations. Markers follow Figure 4 in the main paper; dashed lines mark $0.68\times$ and -1 pp.

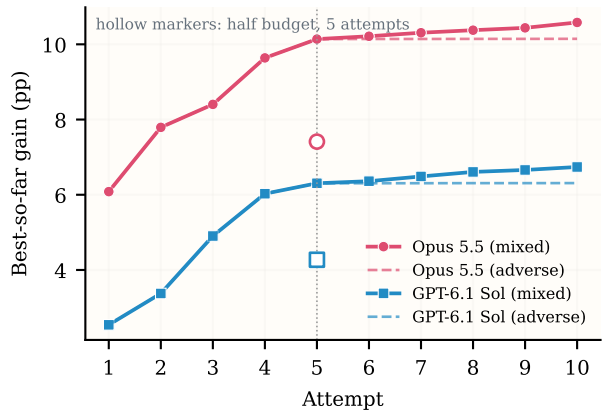


Figure A6. Best-so-far gain over ten attempts for GPT and Opus with Codex. Hollow markers show the half-budget condition after five attempts.

the token, time and search caps (Figure A6). Attempts 6–10 add less than 0.5 pp to the best-so-far gain, supporting the five-attempt default in this scenario. Halving the budget reduces gains by 2.8 pp for Opus and 2.1 pp for GPT while using 45% less active time. In the adverse scenario, the extra attempts add no gain and halving the budget has almost no effect.

Generated-data audit. Valid core high-effort attempts have a 91.1% gate acceptance rate in every feedback view. The other audit values are authored assumptions: binding permutation accounts for 12% of the operation mix with scores only and 31% with evidence regions; new sources per attempt rise from 2.1 to 3.3. Assumed near-duplicate rates are 2.2–2.4%, with 0.6–0.8 contamination blocks per thousand; screening-stage denominators are unspecified. The planned human audit of 260 records by two reviewers is not simulated; findings will be reported after completion.

N. Simulation, Reproducibility and Release

With seed 20261001, we simulate assumed model and harness effects rather than fit product estimates. Scores may rise, plateau or peak early and then decline; failures consume resources. Integer endpoint counts partition each 600-item domain. Alongside the mixed scenario, a null-feedback scenario removes information effects and a region-harm scenario makes region increments negative. All three use the same reporting code. Every performance record carries `SIMULATED_NOT_EMPIRICAL`; SDs describe variation across synthetic repeats, and no p-value is claimed.

In the mixed scenario, initial-only and wrong-region controls score 1.2–1.8 pp below full feedback. Under null feedback, the controls range from 0.38 pp below to 0.19 pp above full feedback. Under region harm, the wrong-region control scores $+0.59$ pp above full feedback for GPT and $+0.06$ pp above for Opus. The assumed effects need not favor full feedback. The base plan has 8,760 training jobs and 10,611 evaluations; with Section M, the ceilings are 9,432 training jobs, 12,099 evaluations and at most 7.2 million answer calls. These are planned limits, not completed jobs.

Empirical runs require pilots to verify compatible endpoints, image and tool delivery, source licenses, item eligibility, image transforms and scorers. Annotation checks sample up to 20 items per task without favoring those with region annotations; two auditors assess relevance, containment and readability. Training pilots confirm common weights, fresh adapters and language-only adaptation. Deliberate failures test forbidden paths, cache leaks, attempt accounting and missing reruns. Empirical records use a new namespace. Their results may contradict these simulations; they will not be adjusted to match the simulated outcomes.

The supplementary package contains the Researcher contract and initial-only override, evaluator-only control contract, DataPlan schema and example, and operation and effort-binding contracts. It also includes the chart renderer with its fixtures and matched-pair builder, simulation generators, all per-attempt and per-trajectory records with content hashes, and scripts that produce result tables and plots. Every reported table value can be regenerated from these records.

Limitations and broader impact. We reuse one evaluation collection per domain for selection and reporting, so same-set gains include adaptation to that collection. Transfer and general tests help assess this limitation but do not eliminate it. Online discovery depends on source licenses, robots policies and contamination screening, which the simulator only records as fields. A Researcher with web access could retrieve benchmark material; the gateway is designed to block it, but pilots must verify that it does so. Records of active hours and tokens by phase help audit the planned cost of empirical work before execution.